

PROJECT REPORT

Entitled

Data Analysis and Performance Insights of a Web Development Company

Submitted by

PRAJAPATI SHYAM KIRTIKUMAR – 2201031000102

in

PARTIAL FULFILLMENT FOR THE DEGREE OF BACHELOR OF TECHNOLOGY

in

INFORMATION TECHNOLOGY



Department of Information Technology

College of Technology

School of Technology, Design and Computer Application

Silver Oak University, Ahmedabad

[April, 2026]



**SILVER OAK
UNIVERSITY**
EDUCATION TO INNOVATION

School of Technology, Design and Computer Application

College of Technology

Department of Computer Engineering

CERTIFICATE

This is to certify that **PRAJAPATI SHYAM KIRTIKUMAR (2201031000102)** has submitted the Project report entitled **Data Analysis and Performance Insights of a Web Development Company** in partial fulfillment of the requirement for the award of the degree in Bachelor of Technology in Information Technology of Silver Oak University, Ahmedabad during the academic year 2025-2026.

Mrs. Parul Sharma
Assistant Professor
Department of Information Technology
College of Technology
Silver Oak University

Dr. Vikas Tulshyan
Head of the Department
Department of Information Technology
College of Technology
Silver Oak University



**SILVER OAK
UNIVERSITY**
EDUCATION TO INNOVATION

DECLARATION

I hereby declare that the Project report submitted along with the Project entitled **Data Analysis and Performance Insights of a Web Development Company** submitted in partial fulfillment for the Bachelor of Technology in Information Technology to Silver Oak University, Ahmedabad, is a Bonafide record of original work carried out by me at Silver Oak College of Engineering and Technology under the supervision of Mrs. Parul Sharma

I further declare that the report does not contain any material previously submitted for the award of any degree or diploma in this or any other university. It is also certified that no part of this report has been copied from any source without proper acknowledgment, and all referenced materials, figures, and information from other sources have been appropriately cited in the report.

Name of student

Sign of student

Prajapati Shyam Kirtikumar

ACKNOWLEDGEMENT

We would like to express our sincere gratitude to our esteemed institution, **Silver Oak University**, and the **College of Technology**, for providing us with the opportunity and necessary resources to undertake this 8th Semester Project Report. We are deeply thankful to our internal guide, **Mrs. Parul Sharma**, for their continuous guidance, valuable suggestions, and constant encouragement throughout the development of our project titled “Data Analysis and Performance Insights of a Web Development Company”. Their support and expertise played a crucial role in shaping our ideas and successfully completing this project.

We would also like to extend our heartfelt thanks to **Dr. Vikas Tulshyan**, Head of the Information Technology Department, for his support and for creating an environment that promotes innovation, learning, and practical knowledge.

Finally, we express our sincere gratitude to our parents, family, and friends for their unwavering support, encouragement, and understanding throughout the completion of this project.

ज्ञानं परमं भूषणम्

ABSTRACT

In today's digital era, web development companies generate a large amount of data related to clients, projects, website performance, user behavior, and business operations. This project focuses on analyzing real-world data provided by a web development company during an internship. The primary aim of this project is to clean, process, and analyze the data to extract meaningful insights that help improve project efficiency, website performance, and business decision-making. Various data analysis techniques such as **data preprocessing, exploratory data analysis (EDA), visualization, and statistical analysis** are used. Tools like **Python, Excel, SQL, and Power BI** are applied to present insights in an understandable format. The outcome of this project helps the company identify performance trends, client behavior patterns, and areas for improvement.



List of Figures

Fig 3.1 System flow Diagram	12
Fig 3.2 Data Flow Diagram	12
Fig 3.3 ER Diagram	13
Fig 3.4 Use Case Diagram	13
Fig 4.1 CSV file dataset	16
Fig 4.2 Revenue by project type Bar chart	16
Fig 4.3 technology distribution pie chart	17
Fig 4.4 Team size vs Project cost scatter plot	17
Fig 4.5 Power BI dashboard Overview	18
Fig 4.6 Power BI dashboard Detailed Analysis	18
Fig 4.7 Power BI dashboard Country wise analysis	19
Fig4.8 Power BI dashboard Technology wise analysis	19
Fig4.9 Power BI dashboard Final Insights	20

List of Tables

Table 1.1: Project Time Estimation	6
Table 1.2: Project Schedule Table	7
Table 5.1: Data Processing Test Cases	23
Table 5.2: Dashboard and Visualization Test Cases	24
Table 6.1: Problem and Solution	26

Table of Contents

Declaration	ii
Acknowledgement	iii
Abstract	iv
List of Figures	v
List of Tables	vi
Table of Contents	vii – viii
Chapter 1 Introduction To Project	1
1.1 Project Summary	1
1.2 Purpose	1
1.3 Objective	2
1.4 Scope	3
1.5 Technology	3
1.6 Literature Review	4
1.7 project Planning	4
1.7.1 Project Development approach and Justification	5
1.7.2 Project Effort and Time, Cost Estimation	6
1.8 Project Scheduling	7
Chapter 2 System Analysis	8
2.1 Study of Current System	8
2.2 Problem and Weakness of Current System	8
2.2.1 Complexity of the Multi-Tiered Slab System	8
2.2.2 The “Invisible Power” Drain	8
2.2.3 Information Fragment in Support	9
2.3 Requirements of New System	9
2.4 System Feasibility	10
2.4.1 Does the System Contribute to the Overall Objectives	10

2.4.2 Technical, Cost and Schedule Constraints	10
2.4.3 Can The System Be Integrated ?	10
Chapter 3 System Design	11
3.1 System Design & Methodology	11
3.1.1 Architectural Overview	11
3.1.2 Component Design (Key Components)	11
3.1.3 Data Modules	12
3.2 Visual Diagrams	13
Chapter 4 Implementation	14
4.1 Implementation Platform	15
4.2 Modules Specification	15
4.3 Results	15
4.4 Result Analysis	20
Chapter 5 Testing	22
5.1 Testing Plan	23
5.2 Test Results and Analysis	24
5.2.1 Test Cases	24
Chapter 6 Conclusion	25
6.1 Overall Analysis of Project	25
6.2 Dates of Continuous Evaluation	25
6.3 Problem Encountered and Solutions	26
6.4 Summary of Project	27
6.5 Limitation and Future Enhancement	27
References	28

CHAPTER 1: INTRODUCTION TO PROJECT

1.1 PROJECT SUMMARY

The project “**Data Analysis and Performance Insights of a Web Development Company**” was developed to analyze the operational and financial performance of a web development company. The project was created using **Python in Jupyter Notebook and dashboard visualization in Power BI**.

The main purpose of the project is to convert raw company data into meaningful insights. The dataset contains information related to projects, employees, clients, project budget, revenue, completion status, project duration, and technologies used.

Initially, the dataset was imported into Jupyter Notebook where data cleaning and preprocessing were performed. Missing values, duplicate records, and incorrect formats were removed using Python libraries such as Pandas and NumPy. After cleaning, the data was analyzed to identify project trends, employee performance, project delays, revenue generation, and client-related insights.

Different visualizations such as bar charts, pie charts, line charts, histograms, and heatmaps were created using Matplotlib and Seaborn.

The cleaned dataset was then imported into Microsoft Power BI to create interactive dashboards. Four dashboard pages were designed: Company Overview, Employee Performance, Revenue Analysis, and Project Analysis. These dashboards help management quickly understand company performance and make better decisions.

1.2 PURPOSE

The main purpose of this project is to analyze the performance of a web development company and transform raw company data into meaningful business insights. The project helps management understand how projects, employees, clients, budget, and revenue affect the overall performance of the company.

- To identify completed, delayed, and pending projects.
- To evaluate employee performance and workload.
- To compare project budget with generated revenue.
- To identify the most profitable projects and clients.
- To analyze monthly revenue trends and company growth.
- To provide an interactive dashboard that supports better decision-making.

1.3 OBJECTIVE

- To collect and analyze the data of a web development company.
- To clean and preprocess the dataset using Jupyter Notebook.
- To identify completed, pending, and delayed projects.
- To analyze employee performance based on assigned projects and productivity.
- To study the relationship between project budget and generated revenue.
- To identify the projects and clients that contribute the highest revenue.
- To analyze monthly revenue trends and overall company growth..

1.4 SCOPE

The project focuses on analyzing the performance and business activities of a web development company using data analysis and visualization tools.

- To analyze project details such as project name, budget, duration, and completion status.
- To study employee performance based on assigned projects and productivity.
- To analyze client-related data and client satisfaction.
- To compare project budget and generated revenue.
- To identify delayed and profitable projects.
- To analyze monthly revenue trends and company growth.
- To create different charts and visual reports in Jupyter Notebook.
- To develop interactive dashboards in Microsoft Power BI.

1.5 TECHNOLOGY

Technology Used: The project was developed using Python-based data analysis tools and dashboard visualization software.

- **Jupyter Notebook** : Used for data cleaning, preprocessing, analysis, and visualization.
- **Python** : Used as the main programming language for the project.
- **Pandas** : Used to read, clean, and manage the dataset.
- **NumPy** : Used for numerical calculations and data handling.
- **Matplotlib** : Used to create charts and graphs.
- **Seaborn** : Used for advanced statistical visualizations.
- **Microsoft Power BI** : Used to create interactive dashboards and reports.
- **Microsoft Excel / CSV File** : Used as the source dataset format.

1.6 LITERATURE REVIEW

Previous studies show that data analysis and business intelligence tools help companies improve productivity, reduce project delays, and make better business decisions.

- Many researchers have found that data analysis is important for understanding company performance and improving decision-making.
- Business intelligence tools such as Microsoft Power BI are widely used to create dashboards and visual reports.
- Studies show that interactive dashboards help management quickly identify project delays, employee workload, and revenue trends.
- Jupyter Notebook is commonly used for data cleaning, preprocessing, and analysis because it supports Python programming and visualization.
- Python libraries such as Pandas and NumPy are useful for handling large datasets and performing calculations.
- Research also shows that Matplotlib and Seaborn help represent data in graphical form, making it easier to understand patterns and trends.
- Researchers have also concluded that analyzing project and employee data can help companies improve efficiency, increase revenue, and reduce project failure.

1.7 PROJECT PLANNING

1.7.1 Project Development Approach And Justification

The project was developed using a step-by-step data analysis approach, starting from data collection to dashboard creation.

process of making this project:

1. First, the company dataset was collected in Excel or CSV format.
2. The dataset contained information related to projects, employees, clients, budget, revenue, project duration, and completion status.
3. The collected data was imported into Jupyter Notebook for preprocessing and analysis.
4. Data cleaning was performed by removing duplicate values, handling missing data, and correcting data formats.
5. After cleaning, the dataset was analyzed using Python libraries such as Pandas and NumPy.
6. Different charts and graphs were created using Matplotlib and Seaborn to understand trends and patterns.
7. The cleaned dataset was then imported into Microsoft Power BI.
8. In Power BI, multiple dashboard pages were created to show project status, employee performance, revenue analysis, and project insights.
9. Finally, the results were studied and useful conclusions were generated.

Justification:

- This approach was selected because it is simple, systematic, and easy to understand.
- Using Jupyter Notebook helps in accurate data cleaning and detailed analysis.
- Microsoft Power BI was chosen because it provides interactive dashboards and easy visualization.
- The combination of Jupyter Notebook and Power BI gives both technical analysis and business-level insights.
- This approach helps management take better decisions and improve company performance.

1.7.2 PROJECT EFFORT AND TIME ESTIMATION

Project Effort and Time Estimation: The project was completed in different phases, and each phase required a specific amount of time and effort.

Table 1.1 : Project Time Estimation

Sr. No.	Activity	Estimated Time
1	Requirement collection and understanding the project	1 Day
2	Dataset collection and preparation	1 Day
3	Data cleaning and preprocessing in Jupyter Notebook	2 Days
4	Data analysis using Python libraries	2 Days
5	Creating charts and graphs	1 Day
6	Importing cleaned data into Microsoft Power BI	1 Day
7	Designing Power BI dashboard pages	2 Days
8	Testing and checking dashboard results	1 Day
9	Preparing report and documentation	2 Days

1.8 PROJECT SCHEDULING

Table 1.2: Project Schedule Table

Weeks	Phase	Activity Description	Deliverable
1	Planning	Requirement analysis and defining project objectives	Project scope and objectives defined
2	Data Collection	Collecting dataset (projects, employees, revenue, etc.)	Raw dataset (Excel/CSV)
3	Data Preprocessing	Data cleaning, handling missing values in Jupyter Notebook	Cleaned dataset
4	Data Analysis	Analyzing project performance, employee workload, revenue trends	Analytical insights
5	Data Visualization	Creating charts and graphs using Python libraries	Visual charts and graphs
6	Dashboard Development	Importing data into Microsoft Power BI and creating dashboards	Interactive Power BI dashboards
7	Testing & Documentation	Testing results and preparing final report	Final validated report

CHAPTER 2: SYSTEM ANALYSIS

2.1 STUDY OF CURRENT SYSTEM

The current system mainly uses Excel or basic tools for storing data without proper analysis. Advanced tools like Jupyter Notebook and Microsoft Power BI are not used effectively. As a result, decision-making is slow and based on assumptions instead of data insights.

Characteristics of the Current System:

- **Manual Data Handling:** Data is stored and managed manually in files like Excel, which increases chances of errors.
- **Lack of Data Analysis:** Advanced analysis is not performed using tools like Jupyter Notebook.
- **Limited Visualization:** There are no proper charts or dashboards for understanding data trends.
- **No Interactive Dashboards:** Tools like Microsoft Power BI are not used for real-time insights.
- **Time-Consuming Process:** Report generation and data handling take more time.
- **Poor Decision-Making:** Decisions are based on assumptions rather than actual data.
- **No Real-Time Tracking:** Project status and performance are not updated instantly.
- **Data Inconsistency:** Data may contain errors due to lack of proper cleaning and validation.

2.2 PROBLEMS AND WEAKNESSES IN CURRENT SYSTEM

2.2.1 Manual Data Handling

The current system mainly depends on manual data handling using basic tools like Excel. This increases the chances of human errors and data inconsistency. It also makes the overall process slow and inefficient.

2.2.2 Lack of Data Analysis

There is no proper use of data analysis tools such as Jupyter Notebook for extracting meaningful insights. Due to this, the company cannot identify important patterns like project delays or employee performance. As a result, decision-making becomes less effective.

2.2.3 Lack of Data Visualization

The system lacks proper visualization and does not use tools like Microsoft Power BI for dashboards. Without visual reports, it is difficult to understand data trends quickly. This reduces the ability of management to take fast and accurate decisions.

2.3 REQUIREMENTS OF NEW SYSTEM

1. Functional Requirements (FR)

1. The system should allow importing of dataset (Excel/CSV) into Jupyter Notebook.
 2. The system should perform data cleaning such as removing duplicates and handling missing values.
 3. The system should analyze project data including project status, budget, and duration.
 4. The system should evaluate employee performance and workload.
 5. The system should generate visualizations like charts and graphs.
 6. The system should allow data integration with Microsoft Power BI.
 7. The system should provide interactive dashboards for better understanding of data.
 8. The system should display key performance indicators (KPIs) such as total projects, revenue, and completion rate.
-

2. Non-Functional Requirements (NFR)

1. The system should be user-friendly and easy to understand.
2. The system should provide accurate and reliable results.
3. The system should process data efficiently within less time.
4. The system should be scalable to handle large datasets.
5. The system should ensure data security and prevent data loss.
6. The system should provide good performance without lag.
7. The system should be compatible with standard tools like Jupyter Notebook and Microsoft Power BI.
8. The system should support easy maintenance and updates.

2.4 SYSTEM FEASIBILITY

2.4.1 Technical Feasibility

The proposed system is technically feasible because it uses tools like Jupyter Notebook and Microsoft Power BI which are easy to use and widely available. These tools support data analysis, visualization, and dashboard creation efficiently. The system can be implemented using basic computer resources without advanced hardware requirements.

2.4.2 Economic Feasibility

The system is economically feasible as it does not require high investment. Most of the tools used, such as Python and Jupyter Notebook, are free and open-source. Power BI also provides a free version, making the overall development cost low and affordable.

2.4.3 Operational Feasibility

The system is operationally feasible because it is user-friendly and easy to understand. Users can easily interact with dashboards and analyze data without requiring deep technical knowledge. It helps management in quick decision-making and improves overall company performance.

CHAPTER 3: SYSTEM DESIGN

3.1 SYSTEM DESIGN AND METHODOLOGY

The system is designed using a structured approach that includes data collection, processing, and visualization. The dataset is analyzed using Jupyter Notebook to generate meaningful insights. Finally, interactive dashboards are created in Microsoft Power BI to support effective decision-making.

3.1.1 Data Collection

The first step in the system design is collecting the dataset related to the web development company. The data includes project details, employee information, client data, budget, revenue, and project status. This data is collected in Excel or CSV format for further processing.

3.1.2 Data Processing and Analysis

After data collection, the dataset is imported into Jupyter Notebook for cleaning and preprocessing. Missing values, duplicate records, and errors are removed to ensure data accuracy. Then, analysis is performed using Python libraries to identify patterns, trends, and performance insights.

3.1.3 Data Visualization and Dashboard Design

In the final step, the cleaned data is visualized using charts and graphs. The dataset is then imported into Microsoft Power BI to create interactive dashboards. These dashboards help in understanding project performance, employee productivity, and revenue trends for better decision-making.

3.2 VISUAL DIAGRAMS

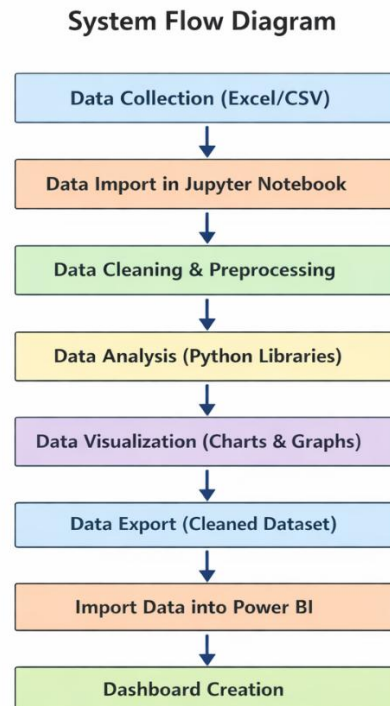


Fig 3.1: System flow Diagram

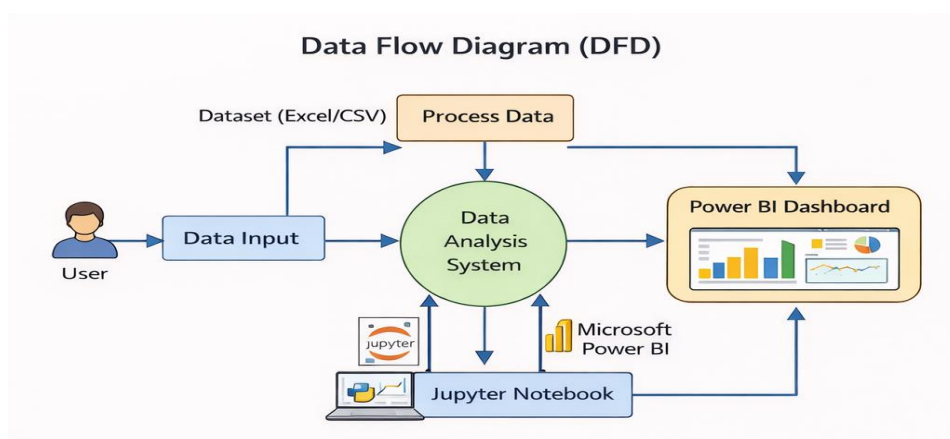
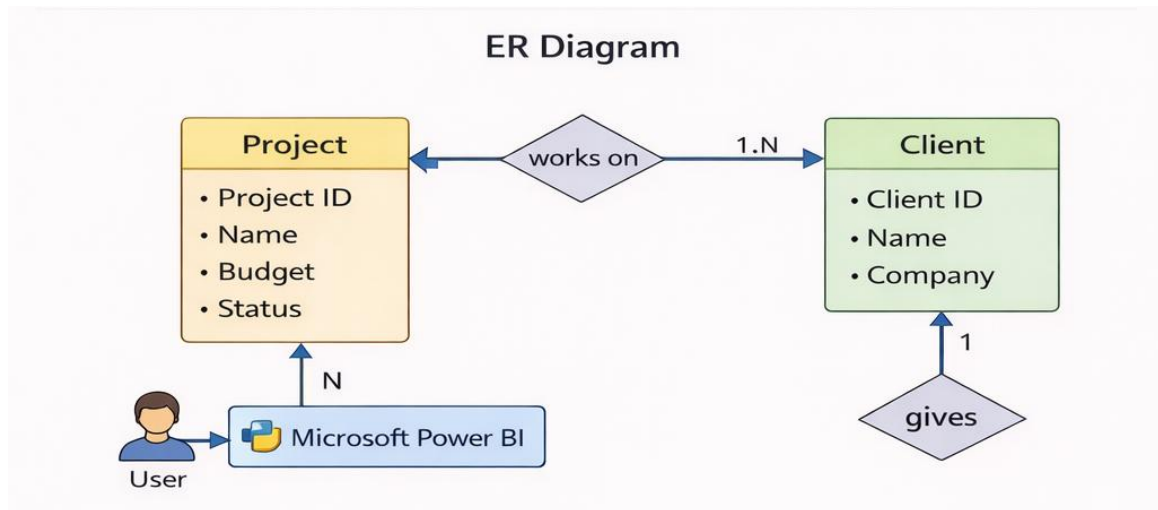
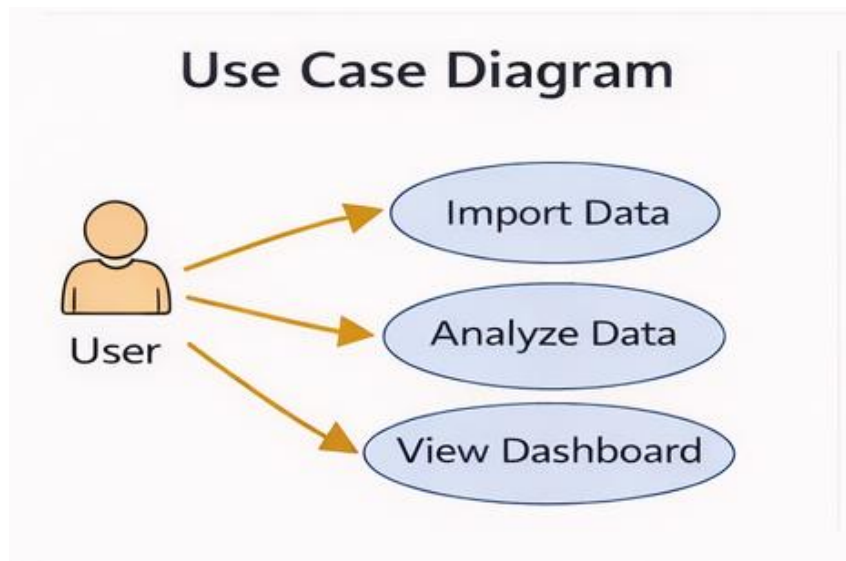


Fig 3.2 Data Flow Diagram

**Fig 3.3 ER Diagram****Fig 3.4 Use Case Diagram**

CHAPTER 4: IMPLEMENTATION

4.1 IMPLEMENTATION PLATFORM

The project was implemented using modern tools for data analysis and visualization. Jupyter Notebook was used for data processing and analysis, while Microsoft Power BI was used for creating dashboards. This combination helps in generating accurate insights and interactive reports.

1. **Software Tools:** Jupyter Notebook for analysis and Microsoft Power BI for dashboard creation.
2. **Programming Language & Libraries:** Python with libraries like Pandas, NumPy, Matplotlib, and Seaborn.
3. **Dataset & System:** Excel/CSV dataset used on a standard computer system without high hardware requirements.

Languages and Frameworks:

- **Programming Language:** Python was used for data analysis and processing.
- **Libraries/Frameworks:** Pandas, NumPy, Matplotlib, and Seaborn were used for data handling and visualization.
- **Tools Used:** Jupyter Notebook for analysis and Power BI for dashboard development.

4.2 MODULES SPECIFICATION

The system is divided into different modules to perform specific tasks efficiently. Each module handles a particular part of the data analysis process using tools like Jupyter Notebook and Microsoft Power BI.

4.2.1 Data Collection Module

This module is responsible for collecting the dataset in Excel or CSV format. It includes project, employee, client, budget, and revenue data required for analysis.

4.2.2 Data Preprocessing Module

In this module, data cleaning is performed using Jupyter Notebook. Missing values are handled, duplicate data is removed, and data is formatted properly for analysis.

4.2.3 Data Analysis Module

This module performs analysis using Python libraries to identify trends such as project performance, employee productivity, and revenue patterns.

4.2.4 Data Visualization Module

In this module, charts and graphs are created using visualization libraries to represent data in an easy-to-understand format.

4.2.5 Dashboard Module

This module uses Power BI to create interactive dashboards. It displays KPIs, project status, revenue insights, and employee performance.

4.2.6 Reporting Module

The final module generates reports and insights from the analysis. These reports help management in decision-making and improving business performance.

4.3 RESULTS

The system was successfully implemented to analyze company data and generate useful insights. The results were obtained using Jupyter Notebook and visualized through dashboards in Microsoft Power BI.

- **Project Performance Analysis:**
The system successfully identified completed, pending, and delayed projects, helping in better project tracking.
- **Employee Performance:**
The analysis clearly showed employee productivity and workload distribution.
- **Revenue Analysis:**
The system identified revenue trends and the most profitable projects.
- **Budget Comparison:**
Budget vs actual revenue analysis helped in understanding financial performance.
- **Data Visualization:**
Interactive dashboards in Microsoft Power BI made data easy to understand.
- **Decision Making:**
The system provided accurate insights which improved overall decision-making.

Detailed Analysis

Count of Project_ID	Project_Type	Client_Country	Technology	Sum of Project_Cost_USD	Sum of Project_Duration_Days	Sum of Team_Size
2	Dynamic Website	Australia	PHP	31825	211	7
2	Dynamic Website	India	Django	45593	176	10
2	Dynamic Website	USA	HTML/CSS/JS	25246	224	12
2	E-commerce	Australia	PHP	26106	149	10
2	Static Website	Canada	Django	44424	192	7
2	Static Website	India	Node.js	8249	97	19
2	Static Website	UK	.NET	15594	186	9
2	Web Application	Australia	.NET	9589	216	18
2	Web Application	India	PHP	23462	149	18
2	Web Application	UK	Angular	32692	271	16
2	Web Application	USA	HTML/CSS/JS	38907	322	12
3	Dynamic Website	Canada	Angular	30202	303	14
3	Dynamic Website	Canada	PHP	54795	212	22
3	Dynamic Website	India	Node.js	25650	258	19
3	Dynamic Website	UK	React	30760	360	21
3	E-commerce	Australia	HTML/CSS/JS	37848	182	12
3	E-commerce	Canada	HTML/CSS/JS	28515	195	20
3	E-commerce	UK	HTML/CSS/JS	58480	360	24
3	E-commerce	USA	Angular	48942	321	14
3	Static Website	Germany	HTML/CSS/JS	32891	364	16
3	Static Website	UK	React	55431	371	26
3	Web Application	Australia	HTML/CSS/JS	51721	318	19
2	Web Application	Australia	Node.js	30700	234	21
1000				12497772	94334	5653

Fig 4.1 CSV file dataset

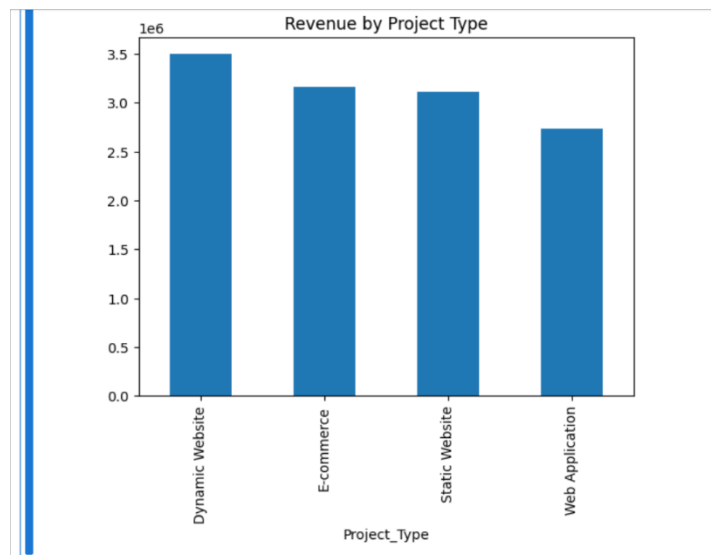


Fig 4.2 Revenue by project type Bar chart

2 Technology Usage (pie chart)

```
[11]: df['Technology'].value_counts().plot(kind='pie', autopct='%1.1f%%')
plt.title("Technology Distribution")
plt.show()
```

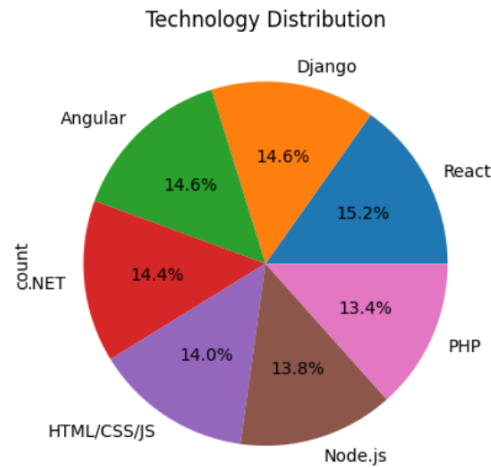


Fig 4.3 technology distribution pie chart

3 Team Size vs Project Cost(scatterplot)

```
[12]: sns.scatterplot(x='Team_Size', y='Project_Cost_USD', data=df)
plt.title("Team Size vs Project Cost")
plt.show()
```

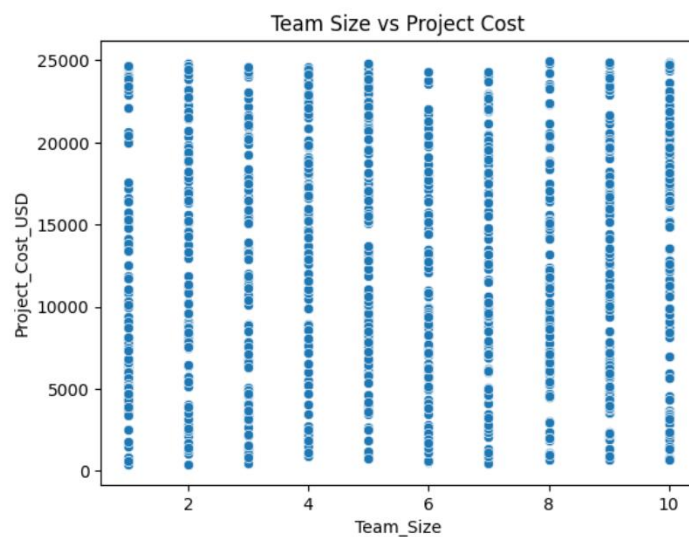


Fig 4.4 Team size vs Project cost scatter plot

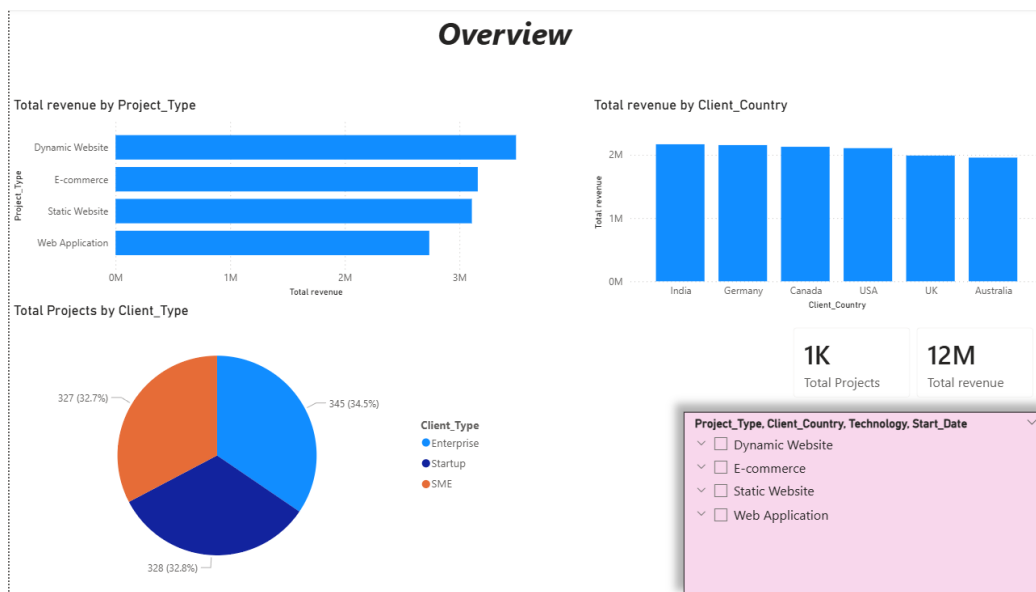


Fig 4.5 Power BI dashboard Overview

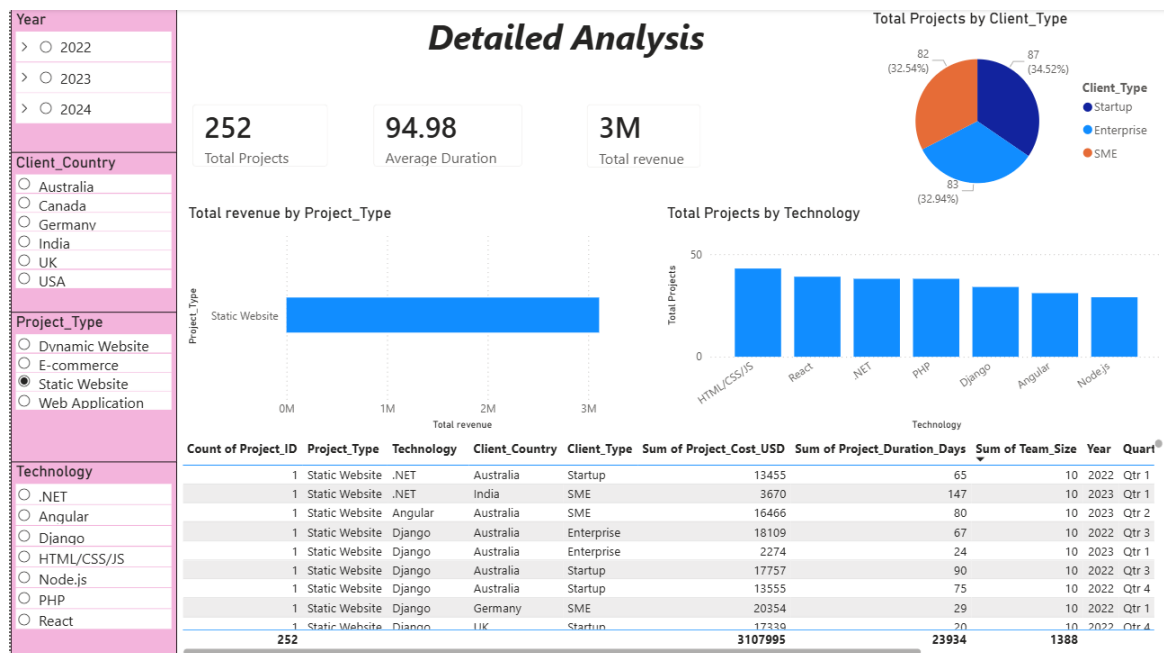


Fig 4.6 Power BI dashboard Detailed Analysis

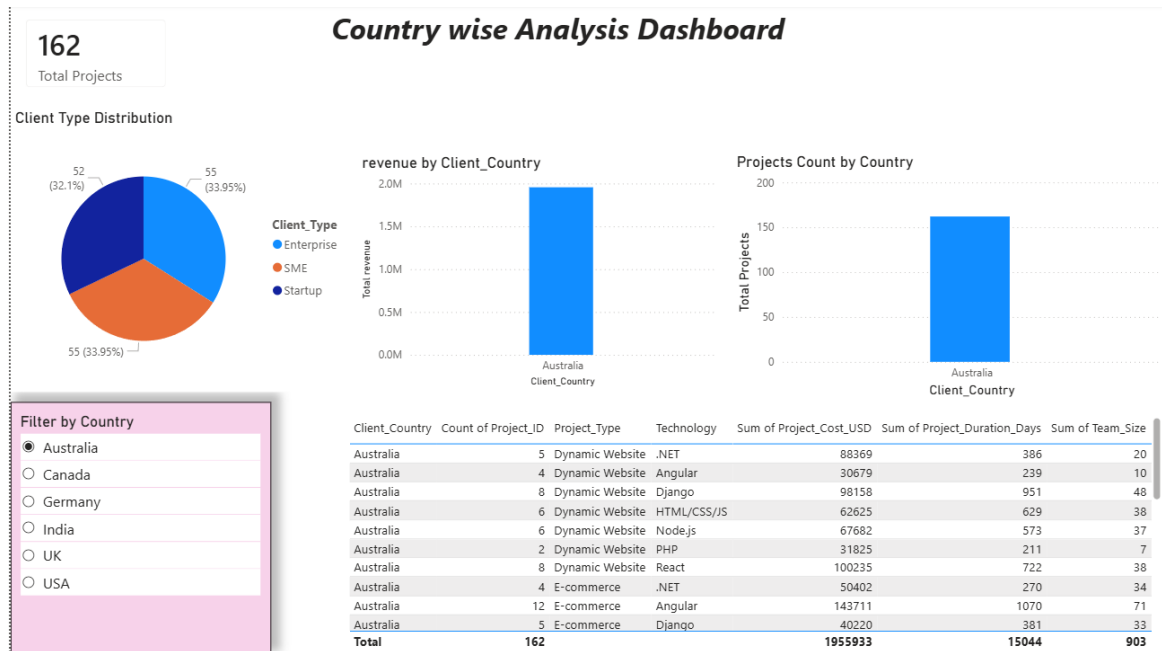


Fig 4.7 Power BI dashboard Country wise analysis

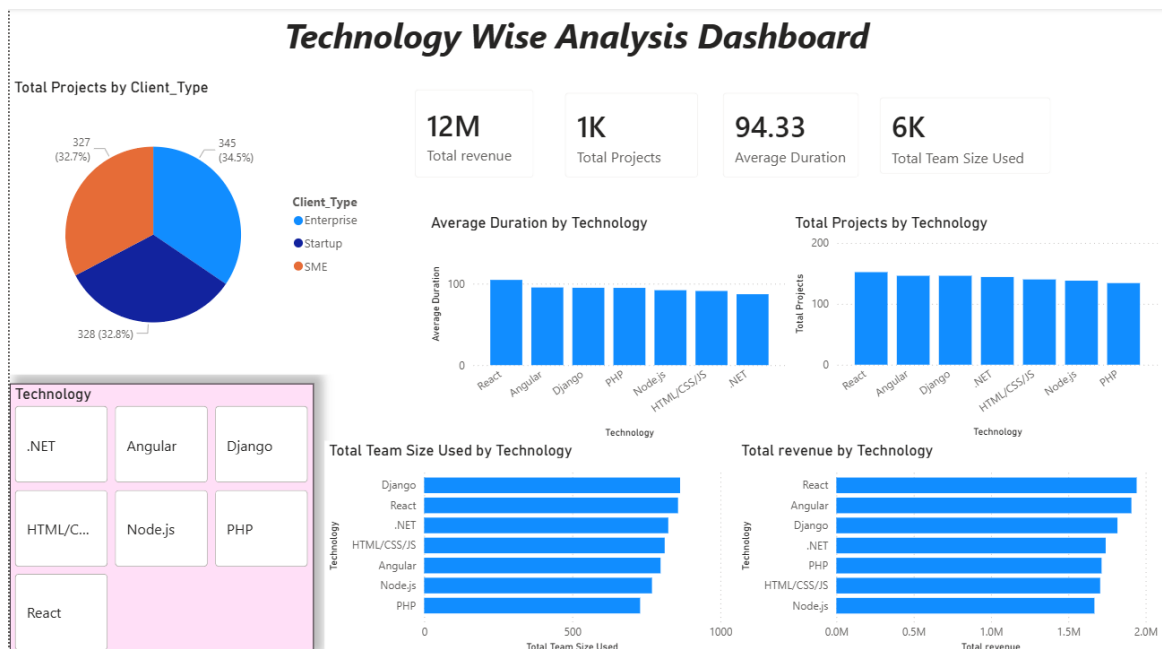


Fig 4.8 Power BI dashboard Technology wise analysis

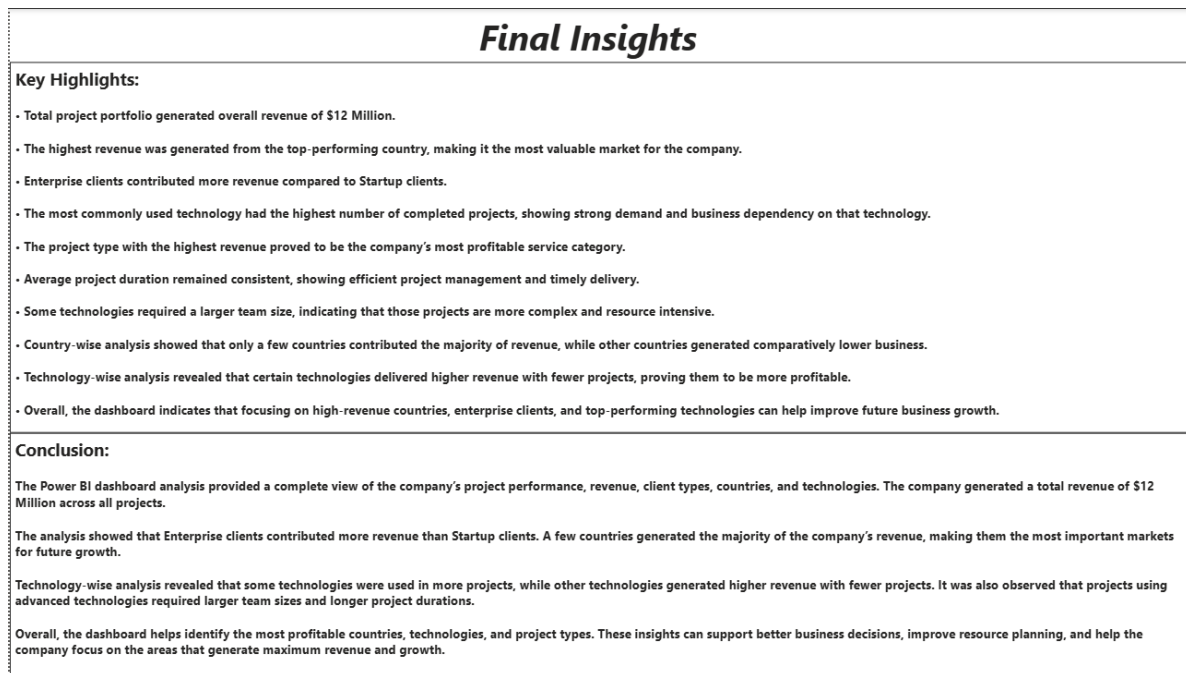


Fig 4.9 Power BI dashboard Final Insights

4.4 RESULT ANALYSIS

The results obtained from the system were deeply analyzed to evaluate the performance and efficiency of the web development company. The analysis helped in identifying key problem areas as well as opportunities for improvement.

- **Project Completion Analysis:**
The system showed that several projects were delayed due to improper planning and time management.
- **Employee Efficiency Analysis:**
Employees handling multiple projects simultaneously showed reduced efficiency and slower completion rates.
- **Workload Distribution Analysis:**
Uneven distribution of work among employees affected overall productivity and caused delays.
- **Revenue Contribution Analysis:**
A limited number of projects contributed to the majority of total revenue, indicating dependency on key projects.

- **Client Analysis:**
Some clients generated more revenue compared to others, helping identify high-value clients.
- **Budget Utilization Analysis:**
Projects that stayed within budget resulted in higher profitability and better resource management.
- **Project Duration Analysis:**
Longer project durations were directly linked to delays and increased costs.
- **Trend Analysis:**
Monthly and yearly trends helped in understanding business growth and seasonal variations.
- **Technology Usage Analysis:**
Certain technologies were more frequently used and contributed to better project outcomes.
- **Data Accuracy and Cleaning Impact:**
Data preprocessing in Jupyter Notebook improved accuracy and reliability of results.
- **Visualization Effectiveness:**
Dashboards created in Microsoft Power BI made it easier to understand complex data.
- **Decision-Making Improvement:**
The overall analysis supported faster and more accurate business decisions.

CHAPTER 5: TESTING

5.1 TESTING PLAN

The testing plan was designed to ensure that the system works correctly and provides accurate results. The system was tested at different stages using Jupyter Notebook and Microsoft Power BI to verify data processing and dashboard outputs.

1. Testing Objectives

1. Data Validation Testing:

Checking whether the input dataset is correct and properly loaded into the system.

2. Data Cleaning Testing:

Verifying that missing values, duplicates, and errors are handled correctly.

3. Functionality Testing:

Ensuring all analysis functions and calculations are working properly.

4. Visualization Testing:

Checking whether charts and graphs are displayed correctly.

5. Dashboard Testing:

Verifying that Power BI dashboards show accurate and interactive results.

6. Performance Testing:

Ensuring the system processes data efficiently without delays.

7. Result Accuracy Testing:

Comparing output results with expected values to ensure correctness.

2. Testing Strategy

The system was tested in different stages including data validation, data cleaning, analysis, and dashboard creation to ensure smooth and error-free performance.

- **Unit Testing:**

Individual modules like data cleaning and analysis were tested separately to ensure each function works correctly.

- **Integration Testing:**

All modules were combined and tested together to verify smooth data flow between Jupyter Notebook and Microsoft Power BI.

- **System Testing:**

The complete system was tested as a whole to ensure it meets all requirements and performs without errors.

5.2 TEST RESULTS AND ANALYSIS

The system was tested at different stages to ensure accuracy, performance, and reliability.

The results confirmed that the system works efficiently and produces correct outputs.

- **Data Accuracy Testing:**

The system produced accurate results after data cleaning and preprocessing in Jupyter Notebook.

- **Functionality Testing:**

All modules such as data collection, analysis, and visualization worked properly without errors.

- **Performance Testing:**

The system processed the dataset efficiently without any major delays or performance issues.

- **Dashboard Testing:**

The dashboards created in Microsoft Power BI displayed correct and interactive results.

- **Integration Testing:**

Data flow between different modules was smooth and error-free.

- **User Satisfaction:**

The system was easy to use and provided clear insights, helping users in better decision-making.

5.2.1 Test Cases

Test Case ID	Test Case Name	Input	Expected Output	Result
TC01	Data Import	CSV/Excel file	Data loaded successfully in Jupyter Notebook	Pass
TC02	Remove Duplicates	Dataset with duplicate rows	Duplicate records removed	Pass
TC03	Handle Missing Values	Dataset with null values	Missing values filled or removed	Pass
TC04	Data Type Conversion	Incorrect data formats	Correct data types assigned	Pass

Table 5.1: Data Processing Test Cases

Test Case ID	Test Case Name	Input	Expected Output	Result
TC05	Data Import in Dashboard	Cleaned dataset	Data loaded in Microsoft Power BI	Pass
TC06	Chart Display	Dataset for charts	Charts displayed correctly	Pass
TC07	KPI Calculation	Project & revenue data	Correct KPI values shown	Pass
TC08	Filter Functionality	User selects filter	Dashboard updates accordingly	Pass
TC09	Report Accuracy	Final dashboard	Accurate and consistent results	Pass

Table 5.2: Dashboard and Visualization Test Cases

CHAPTER 6: CONCLUSION

6.1 OVERALL ANALYSIS OF PROJECT

The project successfully analyzed the performance of a web development company using data analysis techniques. Tools like Jupyter Notebook and Microsoft Power BI helped in generating meaningful insights and visual reports. Overall, the system improved understanding of project performance, employee productivity, and supported better decision-making.

Key Findings:

- **Project Delay:** Project delays mainly occur due to poor time management and excessive workload. Proper planning can reduce these delays.
- **Employee Performance:** Employees handling multiple projects show reduced productivity. Balanced workload improves efficiency.
- **Revenue Contribution:** A small number of projects generate the highest revenue. Focusing on such projects increases profitability.
- **Data Visualization:** Using Microsoft Power BI makes data easy to understand. It helps in quick and effective decision-making.

6.2 DATES OF CONTINUOUS EVALUATION

1 st Internal Review Record				
Presentation Time	From		To	
Presentation Date	Date			

2 nd Internal Review Record				
Presentation Time	From		To	
Presentation Date	Date			

6.3 PROBLEM ENCOUNTERED AND SOLUTIONS

Problem Encountered	Description	Technical Solution
Data Missing Values	Dataset contained null or missing values which affected analysis accuracy	Handled missing values using Pandas functions like fill na()
Duplicate Data	Duplicate records were present in the dataset	Removed duplicates using drop_duplicates() in Jupyter Notebook
Incorrect Data Format	Some columns had wrong data types (e.g., text instead of numbers)	Converted data types using Python functions
Large Dataset Handling	Large data caused slow processing	Optimized data using NumPy and efficient coding techniques
Visualization Issues	Charts were not clear or properly formatted	Improved graphs using Matplotlib and Seaborn

Table 6.1: Problem and Solution

6.4 SUMMARY OF PROJECT

The project focuses on analyzing the performance of a web development company using data analysis techniques. Jupyter Notebook was used for data processing and Microsoft Power BI for visualization. The system helped in generating useful insights for better decision-making.

1. **Project Analysis:** The project analyzed project performance, employee productivity, and revenue trends.
2. **Data Visualization:** Interactive dashboards were created using Microsoft Power BI for easy understanding.
3. **Decision Making:** The system provided accurate insights which improved overall efficiency and business decisions.

6.5 LIMITATIONS AND FUTURE ENHANCEMENT

While the project is fully functional, there are several avenues for further growth as technology evolves.

Current Limitations:

Limited Dataset: The project is based on a limited dataset, which may not represent real-time or large-scale company data.

No Real-Time Data: The system does not use live data, so updates are not reflected instantly.

Basic Analysis Only: Advanced techniques like machine learning are not included for prediction.

Dependency on Tools: The system depends on tools like Jupyter Notebook and Microsoft Power BI.

Future Enhancements:

Real-Time Data Integration: The system can be upgraded to include real-time data for better analysis.

Machine Learning Integration: Prediction models can be added for forecasting project delays and revenue.

Advanced Dashboards: More advanced and dynamic dashboards can be created in Microsoft Power BI.

Automation: Automatic report generation can be implemented to reduce manual work.

REFERENCES

1. Microsoft Power BI Official Documentation
<https://learn.microsoft.com/power-bi/>
2. Power BI Community & Tutorials
<https://community.powerbi.com>
<https://www.w3schools.com/powerbi/>
3. Python Official Documentation
<https://docs.python.org/3/>
4. Jupyter Notebook Official Project
<https://jupyter.org/>
5. NumPy Documentation (Data Analysis Library)
<https://numpy.org/doc/>
6. Pandas Documentation (Data Manipulation Library)
<https://pandas.pydata.org/docs/>
7. Matplotlib Documentation (Data Visualization)
<https://matplotlib.org/stable/contents.html>
8. Kaggle Datasets & Tutorials (for practice data)
<https://www.kaggle.com/>
9. Google Scholar – Research Papers on Data Analysis
<https://scholar.google.com/>